INCH-POUND

MIL-C-71134 (AR)
AMENDMENT 2
26 February 2001
SUPERSEDING
AMENDMENT 1
30 August 1996

MILITARY SPECIFICATION

CARTRIDGE, HE, 120MM, M933 W/FUZE M745
LOADING, ASSEMBLING AND PACKING

This amendment forms a part of MIL-C-71134 (AR), dated 01 July 1992 and is approved for use by the U.S. Army Armament Research, Development and Engineering Center, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 3

2.1.2, Under PRODUCT AND PACKAGING DRAWINGS: Add:

"12961145 - Packing and Marking of Interplant Shipping Pallet Box for Body Loading Assemblies for 120mm Mortar"

PAGE 4

3.3, Delete in its entirety and substitute:

"3.3 Surface cavitation. Cavities formed in Segment D (see 3.7) due to chipped or broken explosive from the sidewall of the fuze well shall not exceed in aggregate more than 1/4 of the circumference. The maximum (max.) dimension of any cavity in the faced (bottom) surface of the fuze well shall not exceed 0.63 centimeters (cm) in any direction and the total area of all cavities shall not exceed 80 percent of the faced surface area. No individual point of explosive shall extend above the flat surface on which the booster assembly would rest. Cavities having a max. Dimension of 0.08 cm or less shall be disregarded in the computation. Disregard any cavitation in the fuze well above Segment D."

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3.7, Delete in its entirety and substitute:

"3.7 Cavitation in high explosive charge.

3.7.1 Projected areas and dimensions. Cavities and cracks within the explosive charge shall not exceed the requirements specified in Table IV when determined as specified in 4.4.3.3 and 4.5.3. Cavities having a maximum projected length of 0.08 cm or less shall be disregarded, provided they are separated by at least 1.27 cm of solid cast.

TABLE IV. Cavitation requirements.

	SEGMENTS (See Figure 1)			
	A	B	C	D
Sum of projected area of the cavities, excluding pipes, cracks and annular rings, square centimeter (sq cm)	0.10	1.61	3.23	3.23
Projected length of any cavity excluding pipes, cracks and annular rings, (cm)	0.32	1.27	1.27	1.90
Piping cavities, max. projected area (sq cm)	0	1.61	3.23	N/A
Piping cavities, max. Projected width (cm)	0	1.27	1.27	N/A
Cracks, max. projected width (cm)	0.08	0.08	0.16	N/A
Annular rings, max. projected width (cm)	0	0	0.63	NA

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Note - If the length of the largest cavity is 0.16 cm or less, the max. total projected area may be 0.32 sq cm.

NA - Not applicable.

3.7.2 Porous areas. Porous areas shall be treated as cavities and shall be subject to the restrictions placed upon the cavities, except that 80 percent of the projected length and 80 percent of the projected area shall be considered.

3.7.3 Cracked charges. Not more that one transverse crack shall be permitted in segment A and not more that a total of three transverse cracks shall be permitted in Segments A, B and C, combines. Cracks shall be disregarded in Segment D.

3.7.4 Annular rings. The presence of annular rings 0.16 cm max. and extending from the shell wall 0.63 cm into the charge shall be disregarded.

3.7.5 Cavities extending into two segments. The entire area of cavities extending into two segments shall be subject to the requirements of both segments.

3.7.6 Base separation. The separation between the base of the projectile and the explosive charge shall not be greater than 0.04 cm.”

3.7.3, First sentence: change "1 1/2 inches" to "38.1 mm" and "1/4 inch" to "6.35 mm”

PAGE 7

4.3.3, Delete “f. Horizontal range probable error is exceeded. (70 degrees F conditioned samples only)” and substitute “f. Horizontal range probable error is exceeded (21 degrees C conditioned samples only)."

PAGE 12

* 4.4.2.2, Delete in its entirety and substitute the included page 12.

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QUALITY CONFORMANCE INSPECTION.

CLASSIFICATION OF CHARACTERISTICS MIL-C-71134 (AR)
AMENDMENT 2

PARAGRAPH 4.4.2.2	TITLE Apadter	SHEET 1 OF 2		DRAWING NUMBER 12577513
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY 12577504 INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Pitch diameter of forward (internal) thread	Level III	3.2	Gage
102	Minor diameter of forward thread	Level III	3.2	Gage
103	Pitch diameter of wear (external) thread	Level III	3.2	Gage
104	Major diameter of rear thread	Level III	3.2	Gage
105	Roundout of forward thread to front face, shoulder and rear pitch diameter	Level III	3.2	Gage
106	Forward thread minimum effective length	Level III	3.2	Gage 1/
107	Distance from rear face to shoulder	Level III	3.2	Gage
108	Failure of protective finish	2/	3.2	Visual/4.5.5
109	True position of adapter holes with front face and rear thread	Level III	3.2	Gage
110	Diameter of adapter holes (2)	Level III	3.2	Gage
111	Depth of adapter holes at full diameter	Level III	3.2	Gage

NOTES:
1/ Minor diameter shall be checked to full form for min. length.
2/ See 4.4.3.4

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PARAGRAPH	TITLE	SHEET 2 OF 2		DRAWING NUMBER
4.4.2.2	Adapter			12577513
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMET PARAGRAPH	NEXT IGHER ASSEMBLY 12577504 / INSPECTION METHOD REFERENCE
112	Depth of adapter holes, max.	Level III	3.2	Gage
113	Diameter of taper at basic lengths (2)	Level III	3.2	Gage
114	Runout of taper with rear tread PD and shoulder	Level III	3.2	Gage
115	Length from front face to shoulder	Level III	3.2	Gage
116	ID rear of internal thread	Level III	3.2	Gage
117	OD of undercut forward of external Thread	Level III	3.2	Gage
118	Surface finish of front face and Shoulder	Level III	3.2	Visual
119	Presence of burrs in threads	Level III	3.10	Visual
120	Phosphate coating missing or inadequate on unpainted surface	Level III	3.2	Visual
121	Width of undercut forward of external thread	Level III	3.2	Gage
Minor				
201	Radii or chamfers incorrect	Level V	3.2	Gage
202	Evidence of poor workmanship	Level V	3.2	Visual

NOTES:

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PAGE 15

4.4.2.5, Special C: Delete “X-Ray” and substitute “4.5.3”

4.4.2.5, Special: change "a", "b", "c" to "A", "B", "C"

Add 4.4.2.5.1 as included page 15a:

Add 4.4.2.5.2 as included page 15b:

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PARAGRAPH	TITLE	SHEET		DRAWING NUMBER
4.4.2.5.1	Packing and Marking of Interplant Shipping Pallet Box for Body Loading Assemblies (Unclosed)	1 OF 1		12961145
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY: Final INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Body Assemblies improperly oriented 1/	Level III	5.4	Visual
Minor				
201	Marking missing, misleading or unidentifiable	Level V	5.5	Visual
202	Evidence of poor workmanship	Level V	3.10	Visual

NOTES: Projectiles must be packed with the fuze end up.

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PARAGRAPH	TITLE	SHEET 1 OF 1		DRAWING NUMBER
4.4.2.5.2	Packing and Marking of Interplant Shipping Pallet Box for Body Loading Assemblies (Closed)			12961145
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY Final INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Contents of box loose	Level III	5.4	Manual
102	Cover not secure	Level III	5.4	Manual
Minor				
201	Evidence of poor workmanship	Level V	3.10	Visual

NOTES:

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PAGE 17

* 4.4.2.7, Critical 1: Delete in its entirety.

PAGE 18

* 4.4.2.7, Add Major defect as follows:

"Major 115, Ignition Cartridge not properly assembled or fully seated … 100% … 3.2 … Gage"

PAGE 26

4.4.3.1.2, Delete "f. Horizontal range probable error is exceeded. (70 degrees F conditioned samples only)" and substitute "f. Horizontal range probable error is exceeded. (21 degrees C conditioned samples only)"

4.4.3.1.3, Delete "f. Horizontal range probable error is exceeded. (70 degrees F conditioned samples only)" and substitute "f. Horizontal range probable error is exceeded. (21 degrees C conditioned samples only)"

PAGE 27

4.4.3.3, Delete in its entirety and substitute:

"4.4.3.3 Cavitation (see 3.7). The sampling plan and procedure for this test shall be accordance with MIL-STD-1235 using CSP-1 plan having the following parameters; i value shall
be 1470, f value shall be 1/10, and the S value shall be 4120. Any cartridge having cavitation in excess of that permitted by the applicable requirements for any segment shall be classed defective. The test shall be performed as specified in 4.5.3 with equipment in accordance with 4.4.4."

PAGE 28

4.5.1, Delete second paragraph in its entirety and substitute:

"The cartridges for first article testing and the first production lot shall be separated into three equal groups: one group of 36 conditioned to 49 ± 1.5 degrees C for 24 hours; one group of 72 conditioned to 21 ± 1.5 degrees C for 24 hours' and one group of 36 conditioned to -32 ± 1.5 degrees C for 24 hours. The quantities shall be tested at charges as indicated in Table III."

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4.5.1, Fourth paragraph, delete first sentence in its entirety and substitute:

"The cartridges shall be fired from a 120mm M121 Carrier-Mounted Mortar at an elevation of 800 mils within two minutes after removal from the conditioning chamber. The 120mm M120 Towed Mortar (ground mount) may be used with prior approval from the Government."

4.5.1, Fourth paragraph, After the third sentence, Add: "The mortar tube shall be dray swabbed every five rounds at Zone 1 and every ten rounds at Zones 0 and 4 counting warmer, spotter, test and reference rounds."

PAGE 29

4.5.1, Change paragraph 3 to read:

"A smear camera shall be used to photograph all test samples. The camera shall be positioned to pick up each round 12.2 meters from the muzzle of the gun … "

4.5.2, Delete in its entirety and substitute the following:

"4.5.2 Strength of adhesive bond. The sample shall be tested for integrity by applying the required load 180 degrees from the bond radially and in a direction away from the axis (center) of the ring."

PAGE 30

4.5.3.1, Change second sentence to read:

"Location of the penetrameter hole shall be 171.45 mm measured from base shoulder of projectile body towards fuze nose threads."

Change seventh sentence to read:

"Penetrameter thickness shall be 0.51 plus or minus 0.25 mm."

4.5.3.1, Seventh sentence: change ".020" to "0.508 mm" and change, "0.10 inches" to ".254 mm".

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PAGE 31

4.5.5, Delete " … unpainted surfaces and areas within 3/16 inch of unpainted surfaces may be protected." And substitute " … unpainted surfaces and areas within 4.76 mm of unpainted surfaces may be protected."

Delete a. in its entirety and substitute: "a. The paint shows more than 3.18 mm creepage, blistering or loss of adhesion of paint from scribe mark, stamp mark, or edge."

PAGE 32

4.5.5:

Delete c. in its entirety and substitute: "c. Any blister exceeds 1.59 mm average diameter."

Delete d. in its entirety and substitute: "d. Any pit exceeds 3.18 mm average diameter."

Delete e. in its entirety and substitute: "e. There are four (4) or more scattered pits exceeding 1.59 mm average diameter."

Delete f. in its entirety and substitute: "f. There are four (4) or more spots within 645.2 sq. mm."

Delete g. in its entirety and substitute: "g. There are three (3) spots within 645.2 sq. mm in two (2) or more places."

Delete i. in its entirety and substitute: "i. There is evidence of rust under the paint (a brown discoloration showing through the paint) in excess of 6.35 mm in diameter total area."

4.5.6, Delete calculation in its entirety and substitute:

"Density g/cc = $\frac{(1.273)\ \text{Mass in grams}}{\text{Height (cm) x Diameter}^2\ \text{(cm)}}$"

5.1, Change Packaging. to "Packaging (M934 ctg.)"

5.2, Change Packing. to "Packing (M934 ctg.)"

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5.3 and 5.4, Delete in their entirety and substitute:

"5.3 Packaging (Loaded Projectile Bodies).

5.3.1 Level A. Not Applicable.

5.3.2 Level B. Not Applicable.

5.3.3 Level C. Packing for filled bodies shall be in accordance with the requirements of drawing 12961145.

5.4 Packing (Loaded Projectile Bodies).

5.4.1 Level A. Not Applicable.

5.4.2 Level B. Not Applicable.

5.4.3 Level C. Packing for filled bodies shall be in accordance with the requirements of drawing 12961145.

5.5 Marking. Unit packaging for cartridge shall be marked in accordance with the requirements of drawing 12577551. Packing shall be marked in accordance with the requirements of drawing 12577570. Interplant packaging shall be marked in accordance with the requirements of drawing 12961145."

5.6 Unitization (M934 cartridge). Unitization procedure shall be in accordance with drawing 19-48-4116/38A for container, ammunition and components on 4-way entry pallets."

The margins of this amendment are marked with an asterisk, or vertical line to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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